

Slow Addition of Sodium Chloride for Colloidal Crystal Assembly

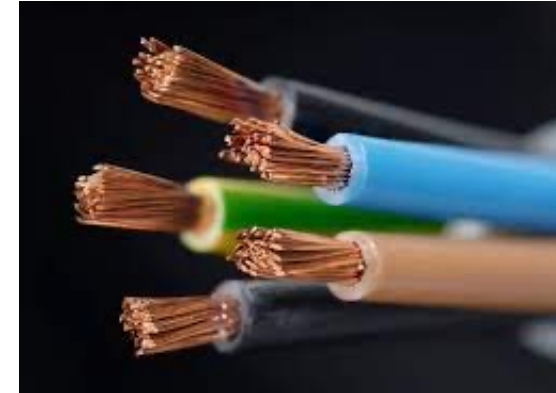
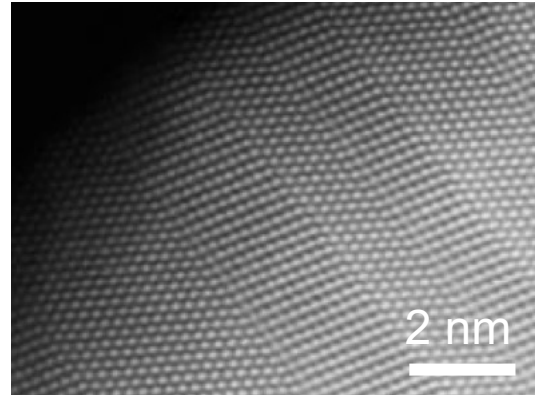
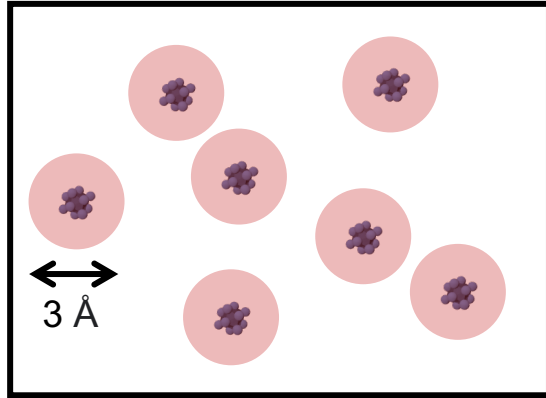
Kaan Borekci

Anisotropic Subgroup

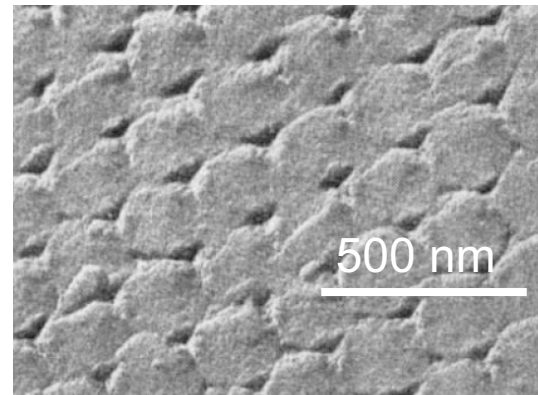
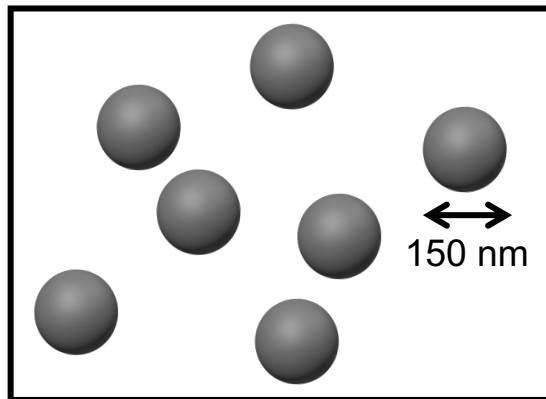
07/12/2023

Classification of Crystals

**Atomic
Crystals**

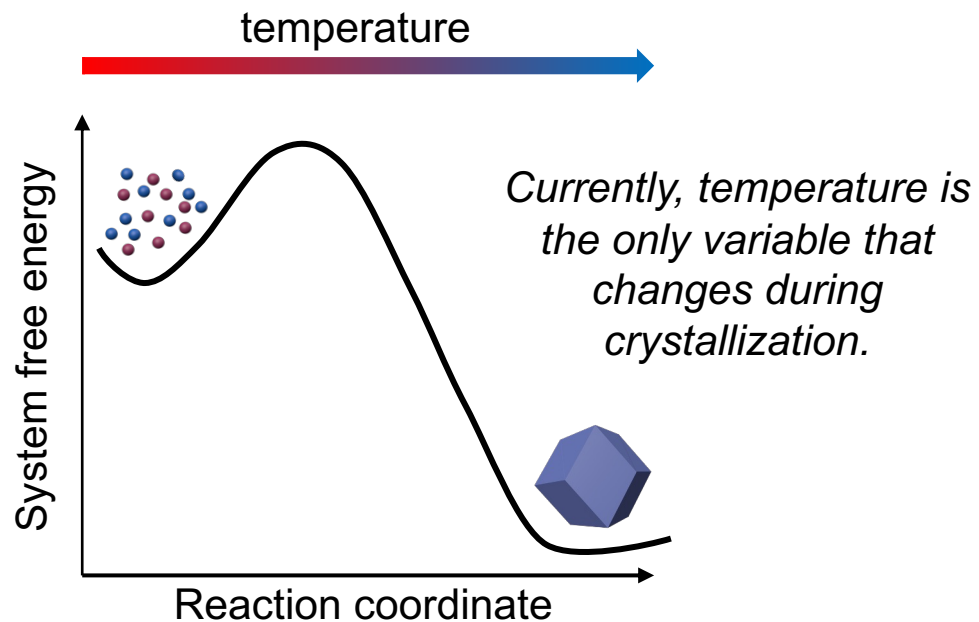
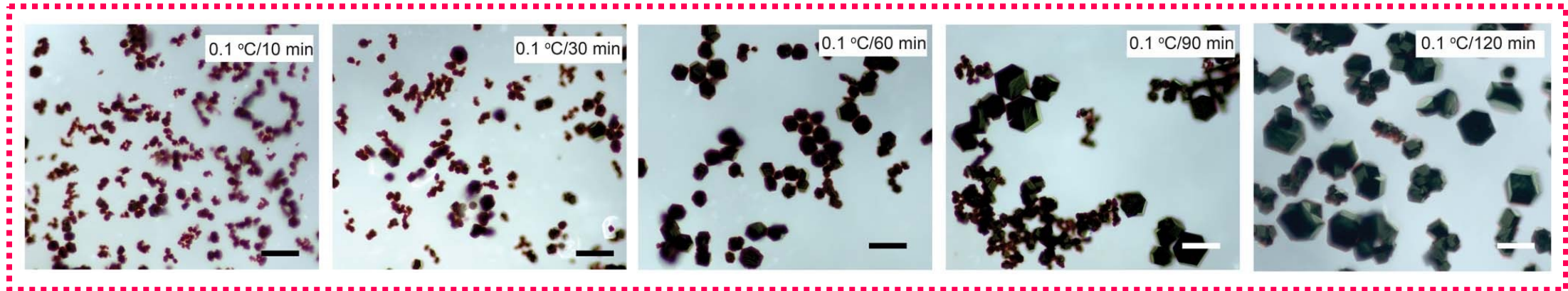


**Colloidal
Crystals**



Design of Crystals

Slower-cooling rate $\longrightarrow$



What if there are other assembly variables that we can manipulate to achieve well-faceted colloidal crystals?

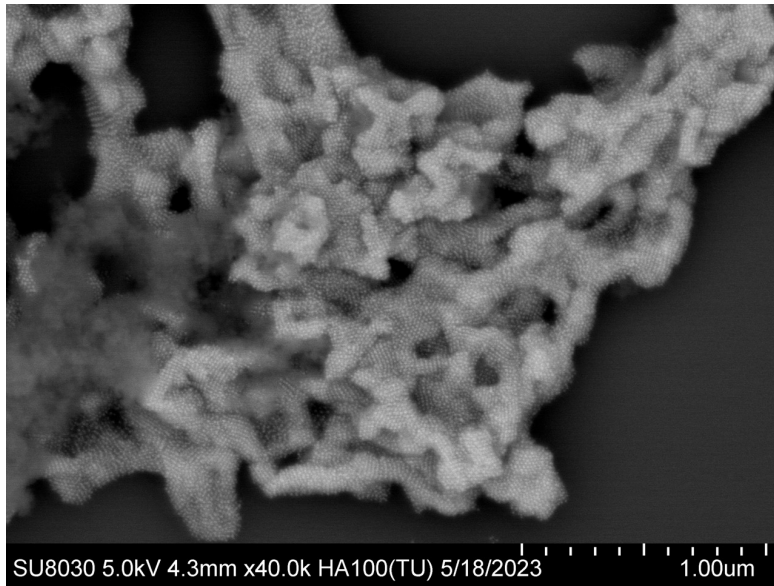
□ [NaCl]

□ [PAE]

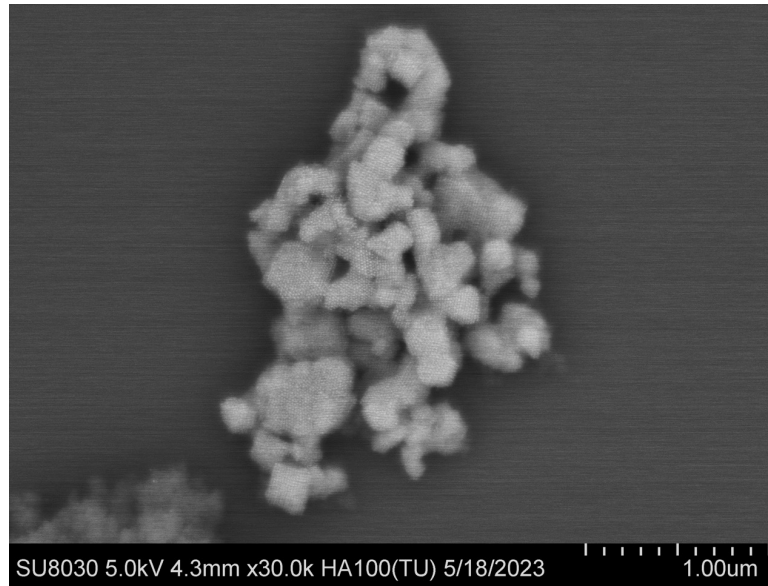
□ [DNA]

Hypothesis 1: Slowly increasing the sodium chloride concentration in a PAE solution over time will create well-faceted colloidal crystals

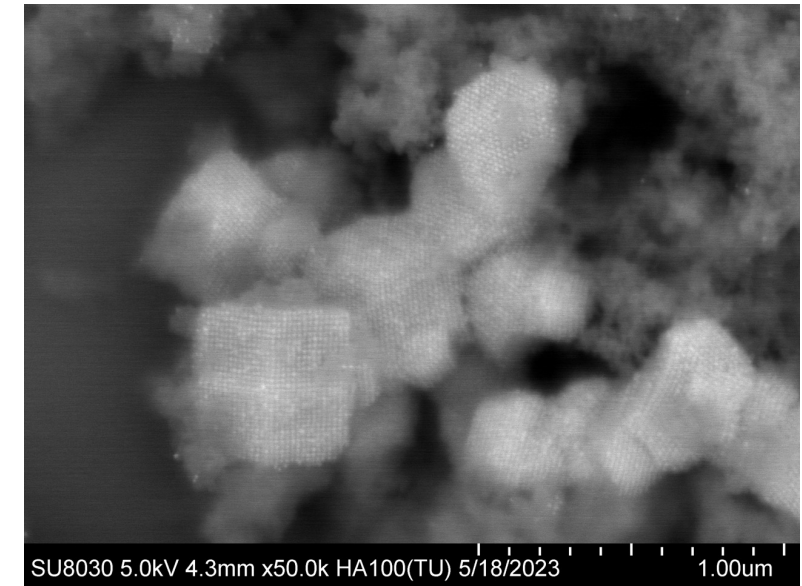
Control (0.5 M NaCl buffer)



Adding 1 M NaCl at 20 μL/hour rate

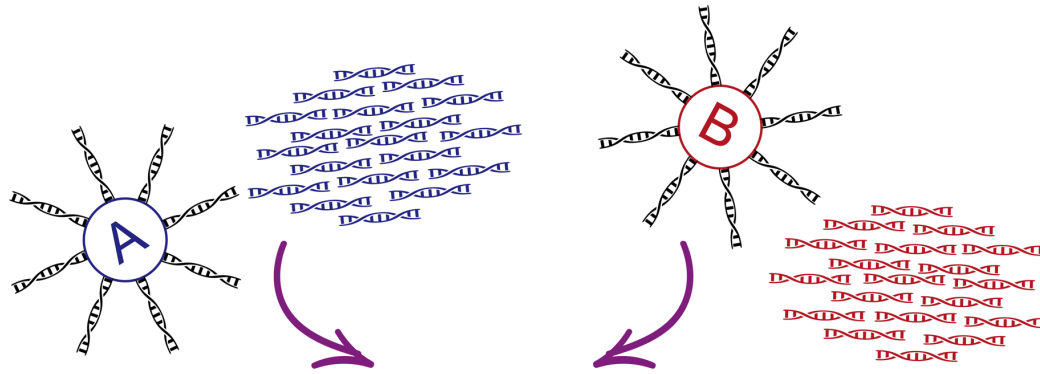


Adding 1 M NaCl at 10 μL/hour rate



Held at 30 °C overnight on platform shaker, syringe pump added to 0.5 M final [NaCl]

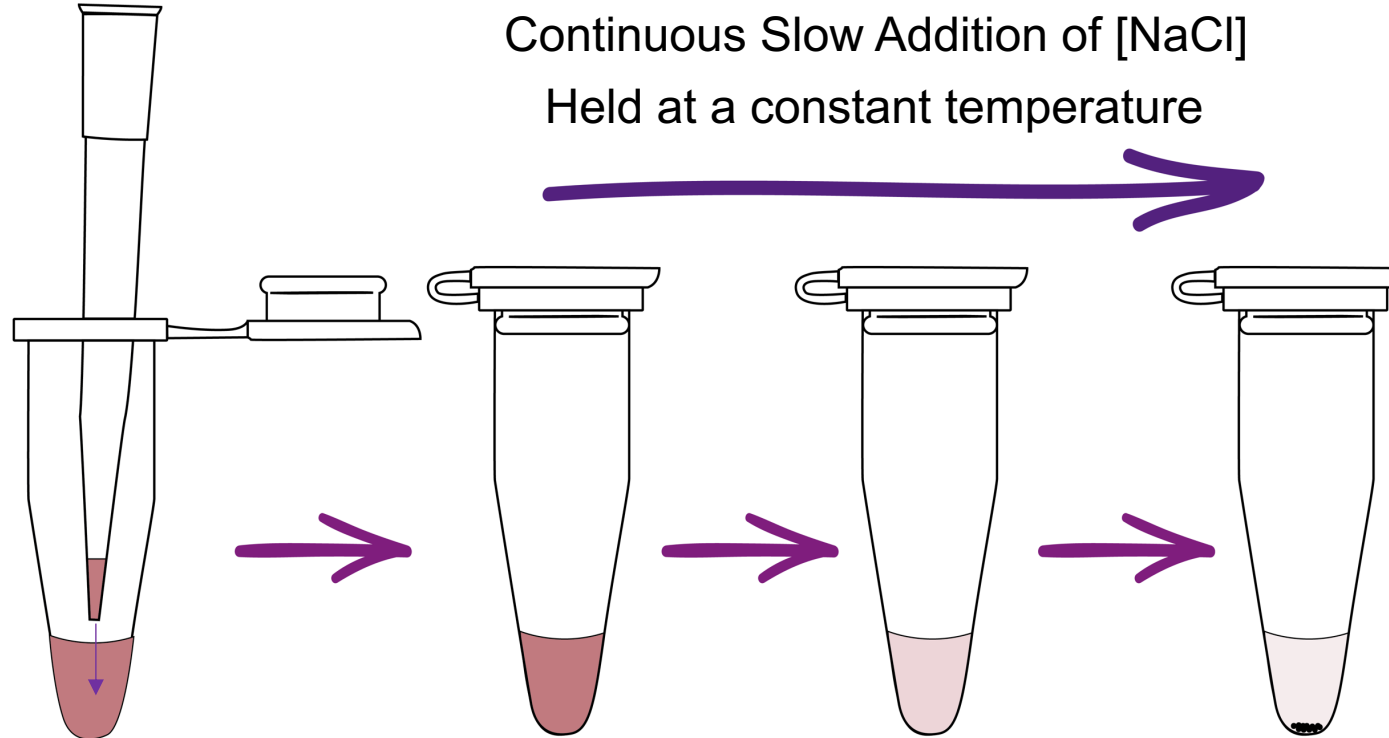
Experimental Set-Up



- 1:1 (PAE A : PAE B)
- 1:600 (PAE : Linker)

To achieve BCC lattice symmetry and RD crystal habit

Continuous Slow Addition of [NaCl]
Held at a constant temperature



Melted

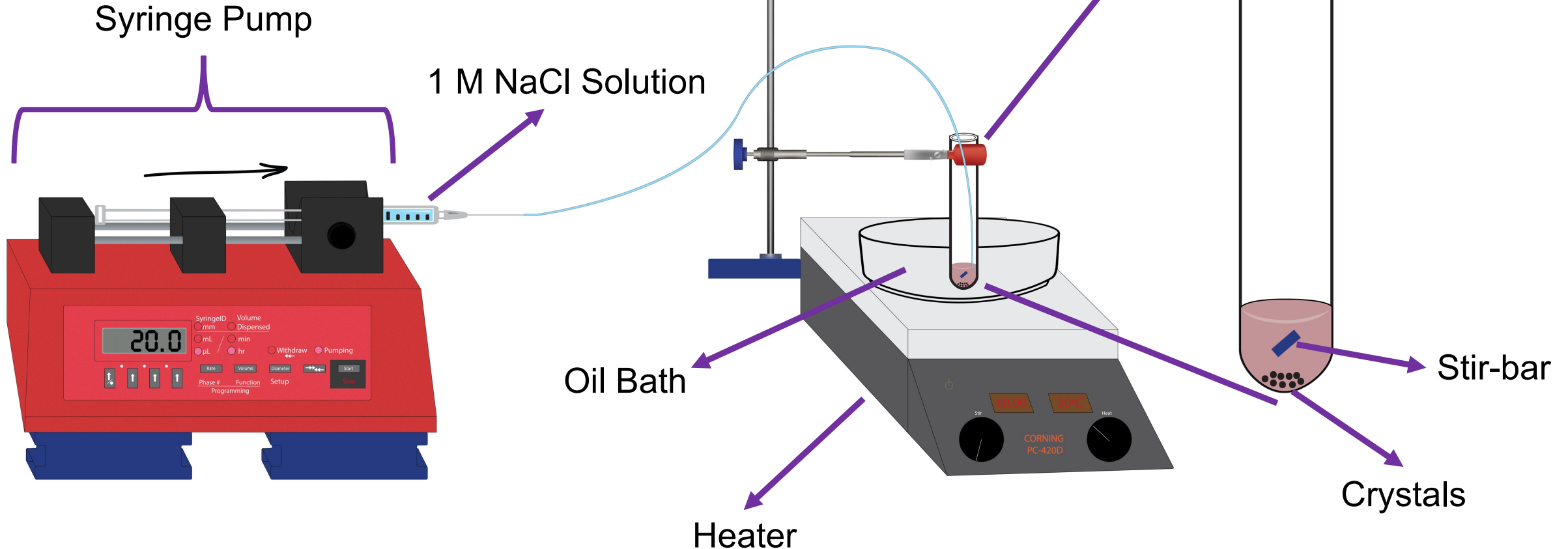
Crystallization

Post-Crystallization

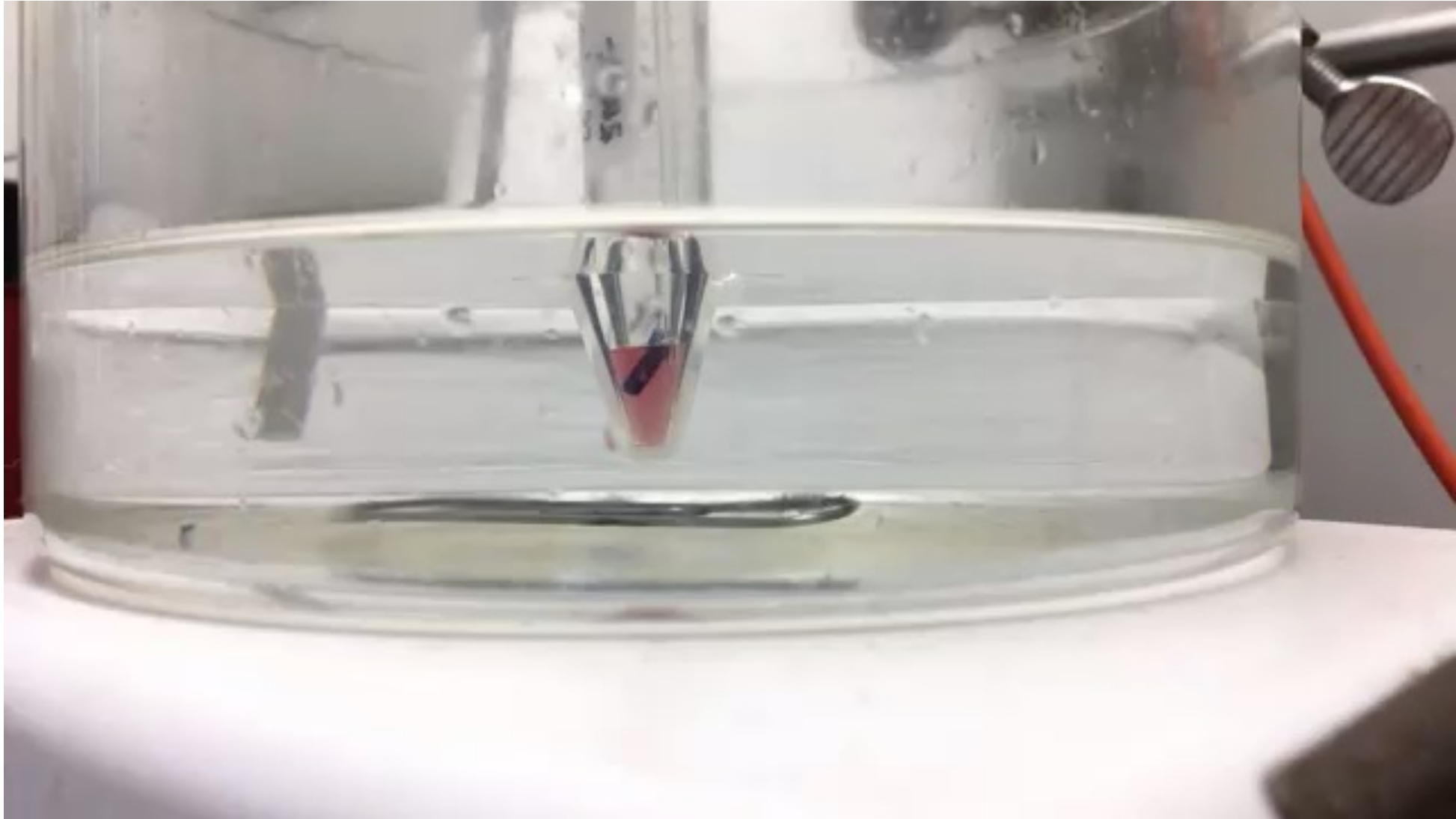
Benchtop Set-Up

Experimental Conditions:

- Volumetric Flow Rate: 20 μ L/hr
- Stir-Bar RPM: 80
- Total Volume Dispersed: 250 μ L
- Temperature of System: 28-30 $^{\circ}$ C



Experimental Set-Up



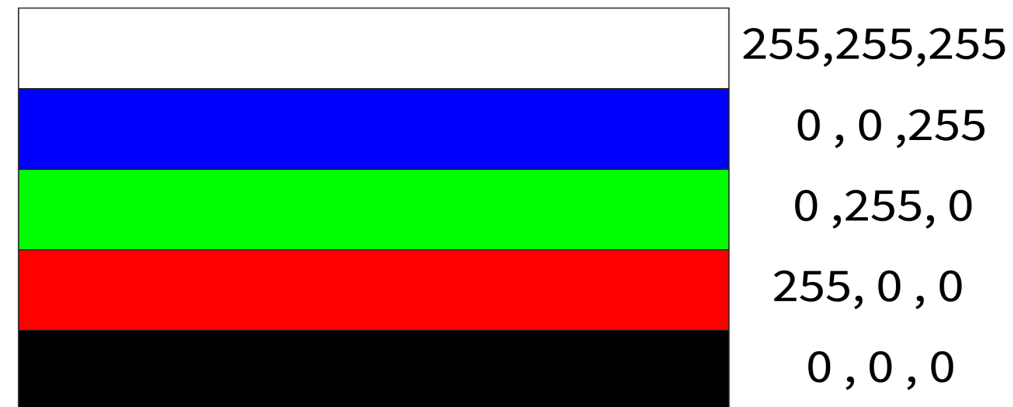
13 hour run, 780 frames, 1 frame = 1 minute

Measuring Pixel Intensity

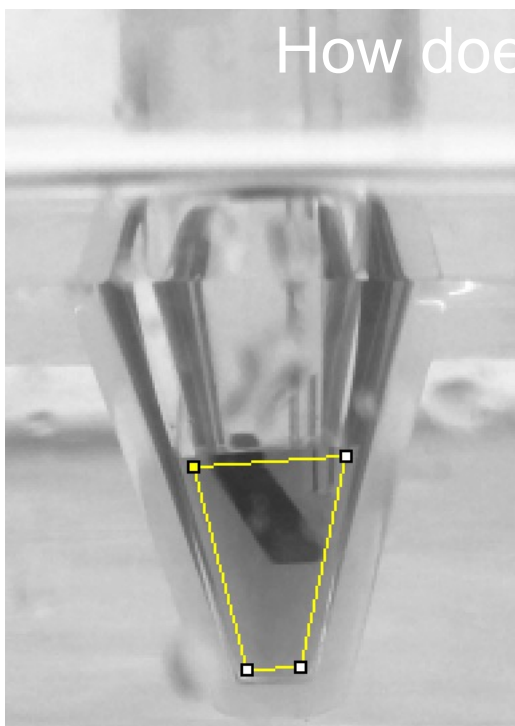
Grayscale: Weighted Average of RGB



RGB

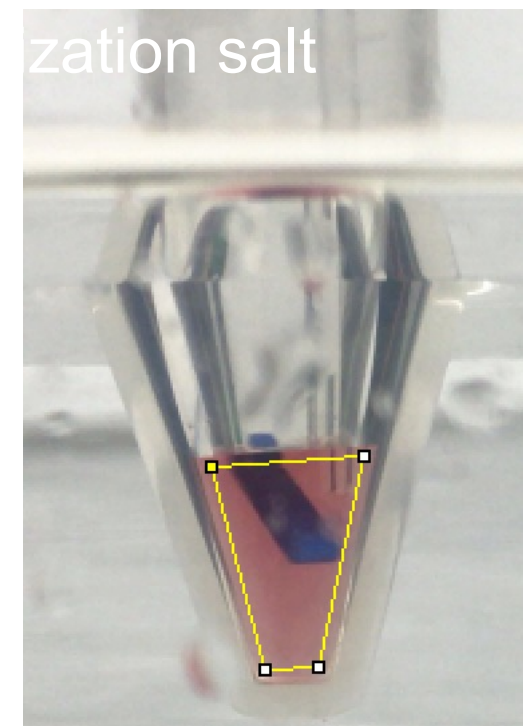


How does



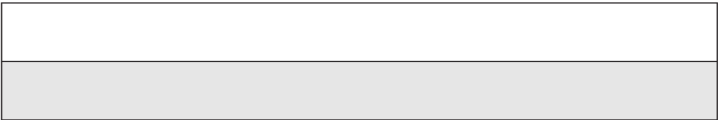
			Results			
	Label	Area	Mean	Min	Max	Perim.
1	Red	11946	114.579	39	169	473.122
2	Green	11946	74.702	28	130	473.122
3	Blue	11946	81.895	40	126	473.122
4	(R+G+B)/3	11946	90.380	37	141	473.122
5	0.299R+0.587G+0.114B	11946	87.461	34	140	473.122

ization salt



Measuring Pixel Intensity

Grayscale: Weighted Average of RGB



255

RGB

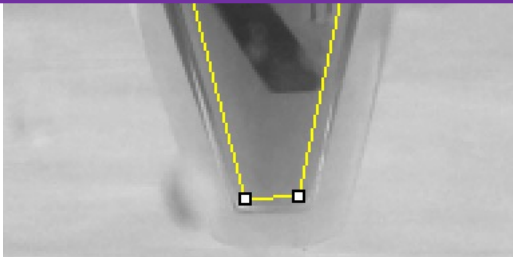


255,255,255

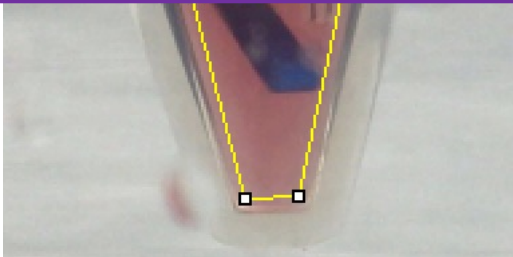
0 0 255

Research Question:

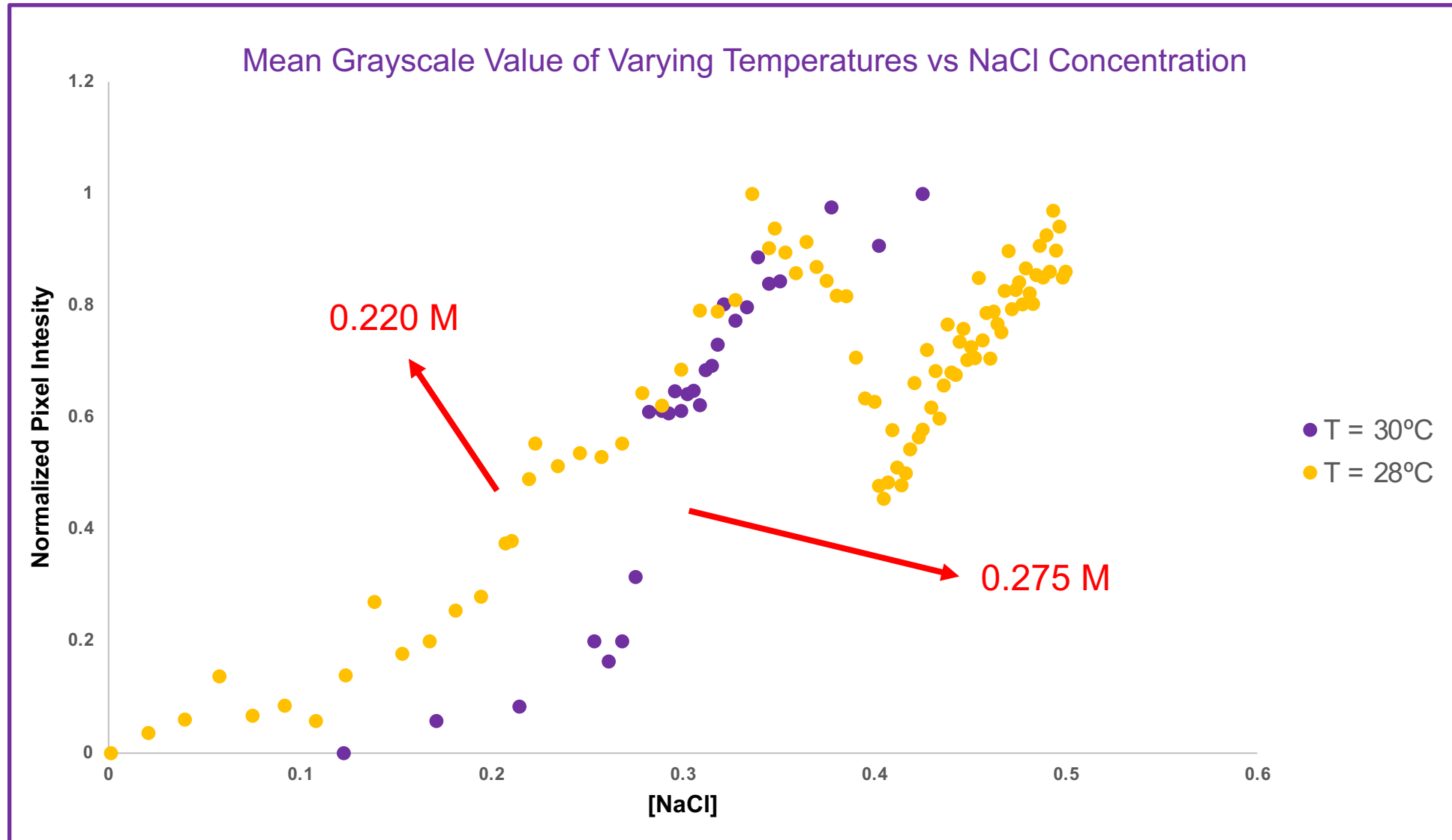
How does the temperature of the medium affect the crystallization salt concentration?



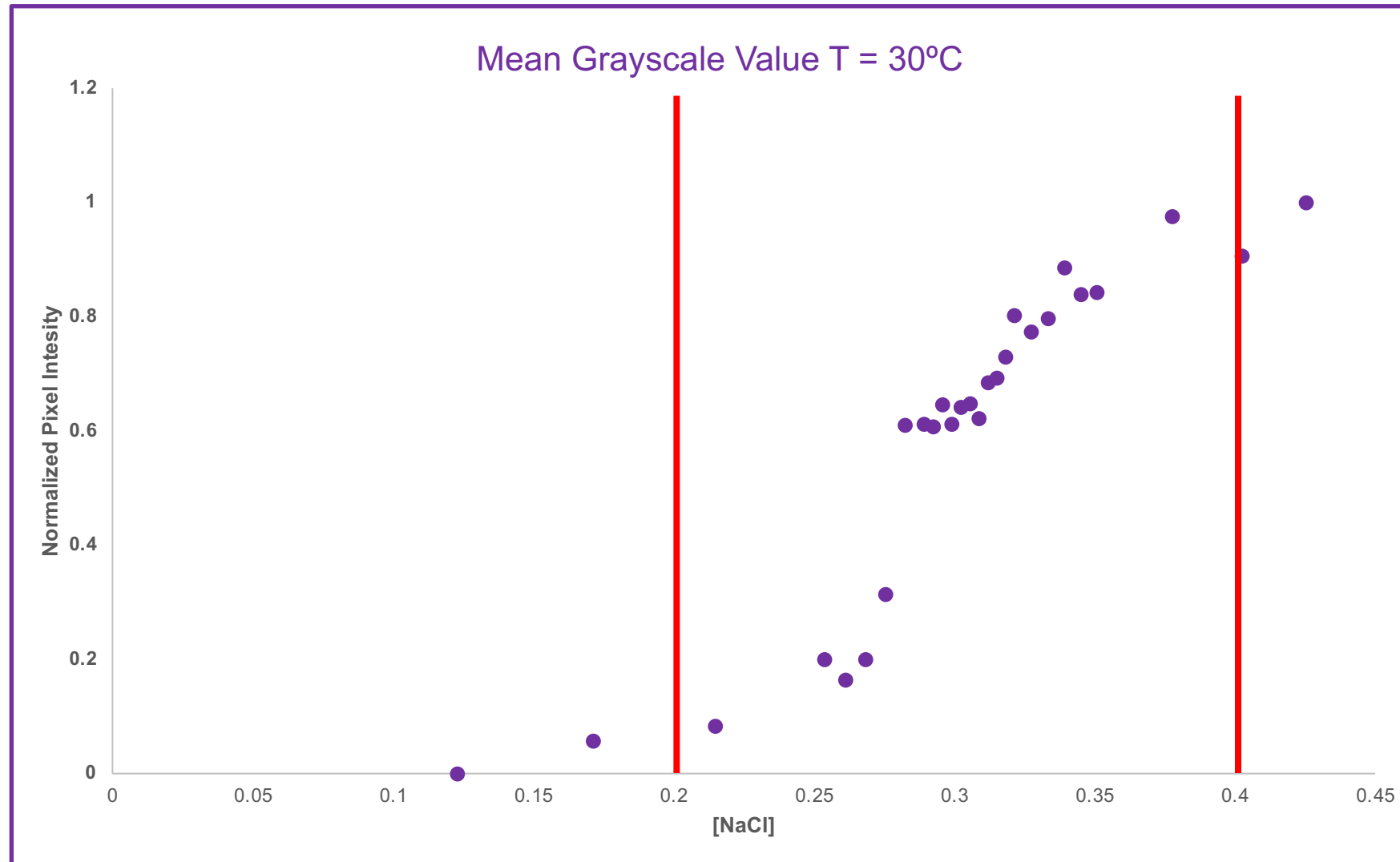
4	$(R+G+B)/3$	11946	90.380	37	141	473.122
5	$0.299R+0.587G+0.114B$	11946	87.461	34	140	473.122



Hypothesis 2: An increase in temperature of the medium increases the salt concentration at which crystallization occurs



Research Question: Can we shorten the salt concentration window?

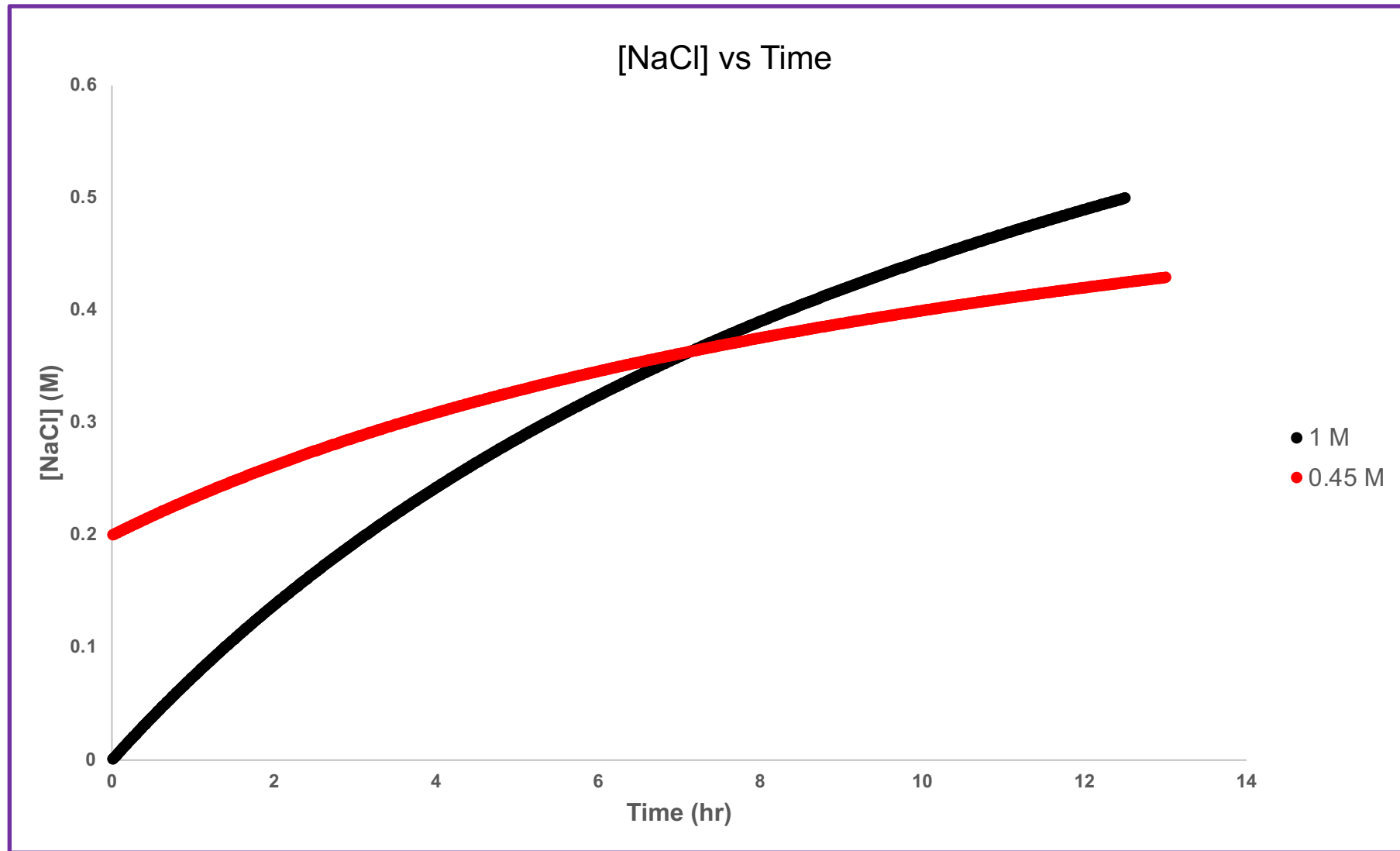


Hypothesis 3: If the concentration window is narrowed, sodium chloride solution can be added at a lower concentration rate over the same experiment time to achieve larger single crystal structures

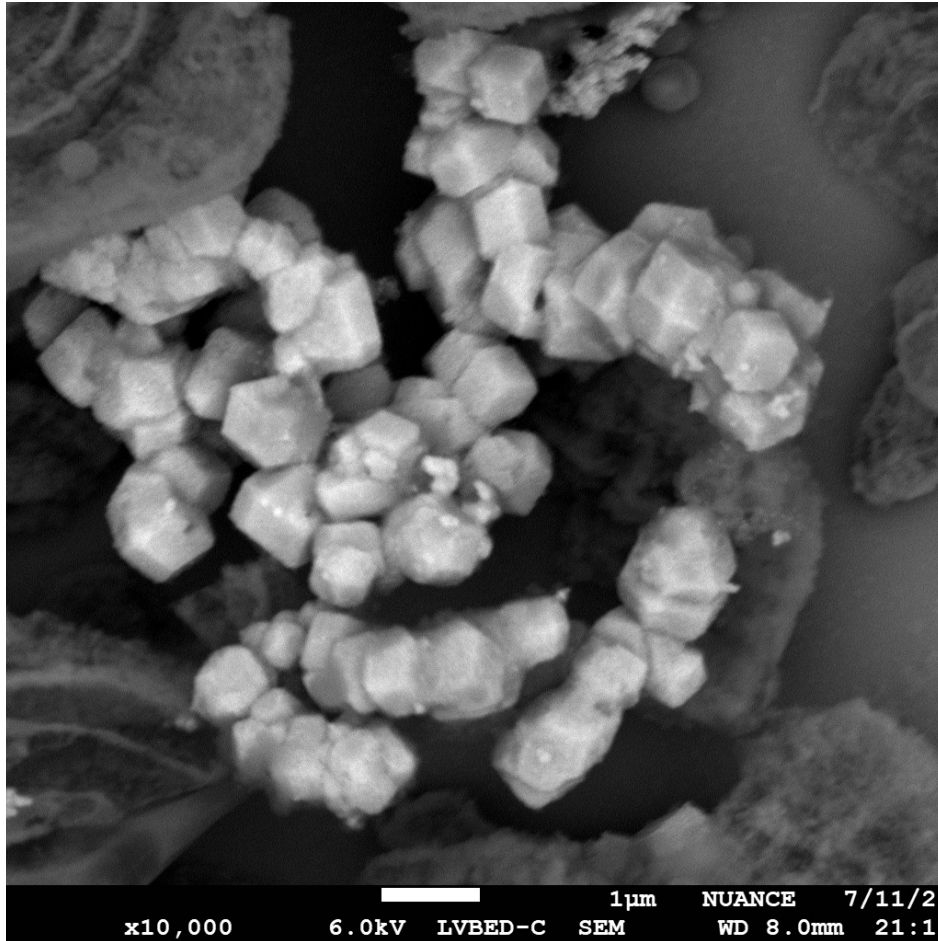


T = 30°C, [NaCl] = 0.45 M, [Solution] = 0.2-0.4 M

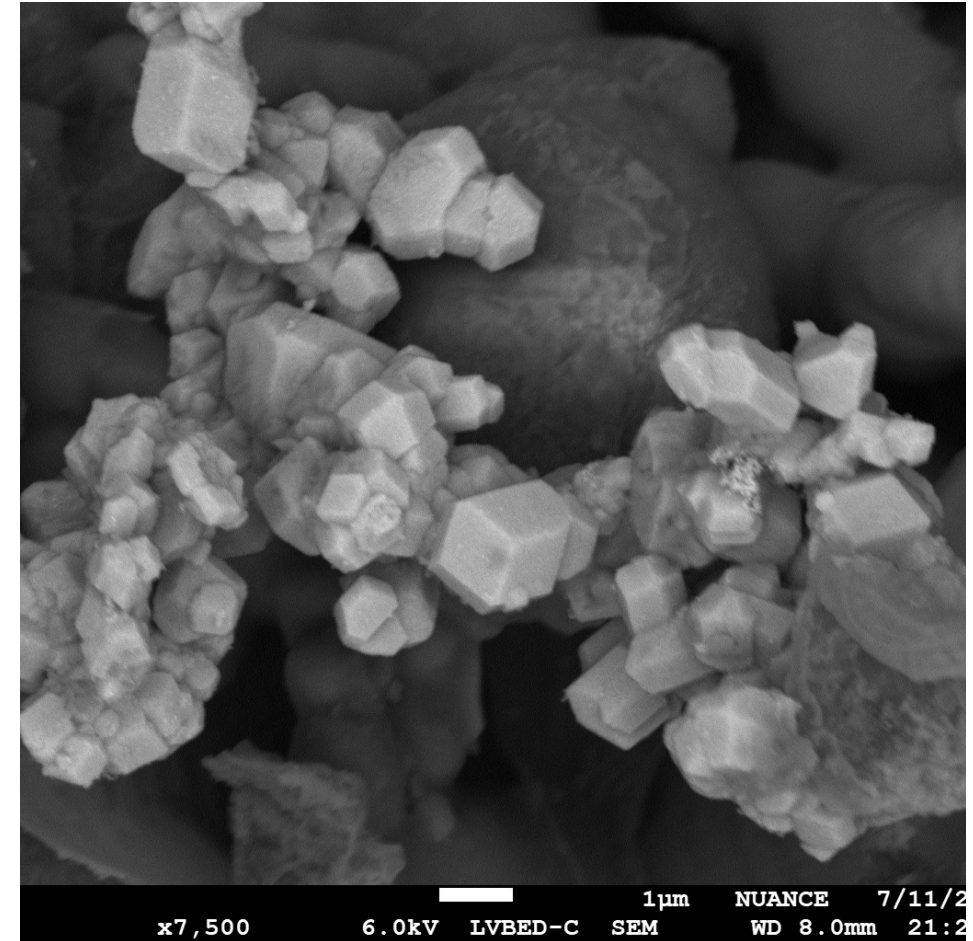
Two different concentration rates were explored in the same experimental time window



Preliminary experiments show slightly larger crystals at slower concentration rate



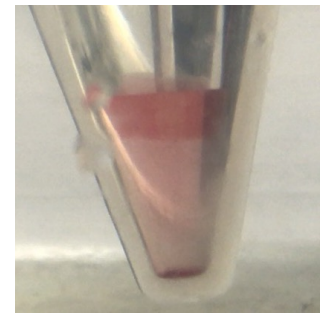
- 1 M NaCl
- Mean Edge Length: 615 nm $\pm$ 0.115
- Number of Crystals Measured: 27



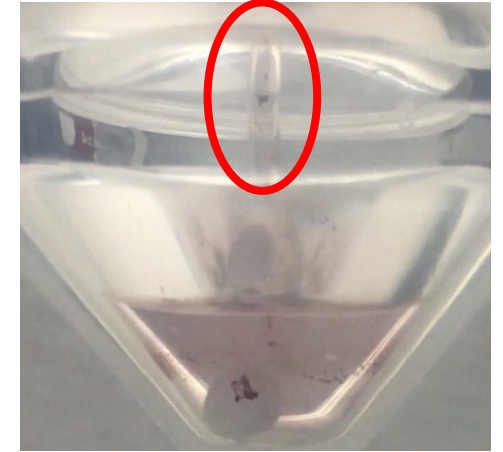
- 0.45 M NaCl
- Mean Edge Length: 835 nm $\pm$ 189
- Number of Crystals Measured: 33

Conclusion

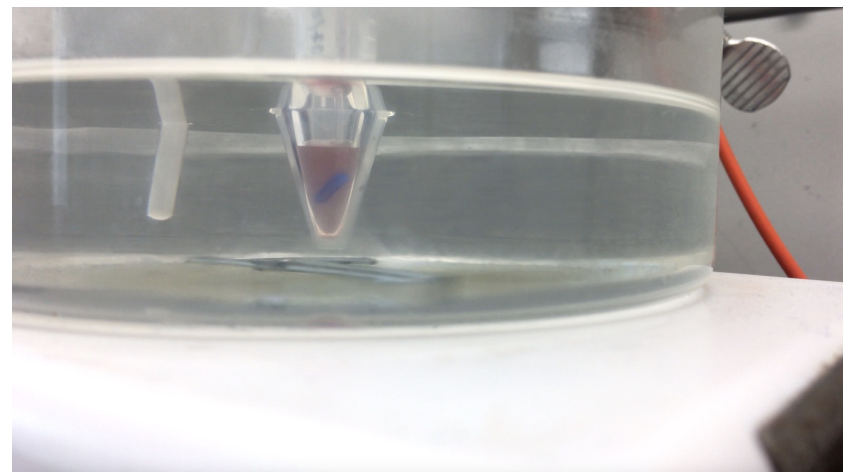
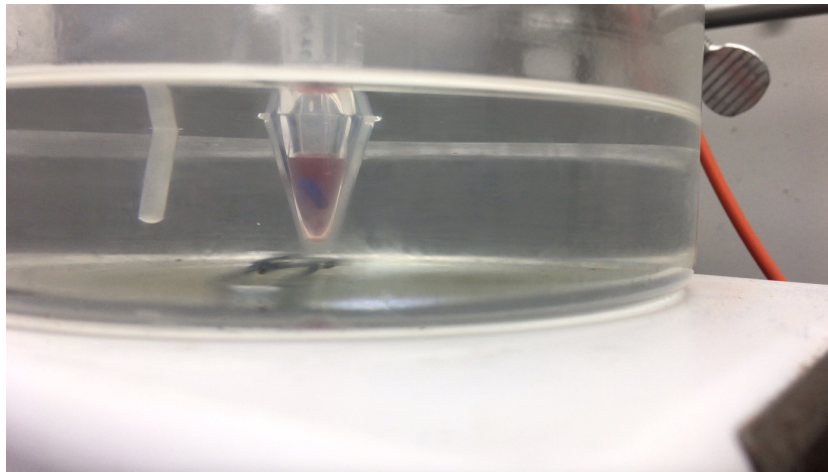
- Slow addition of salt achieved well-faceted crystals
- In initial trials, a phase separation issue was present — which was solved by addition of a stir-bar.
- In some cases, the solution in the test tube backflows into the pipe.
- The surrounding lighting must be consistent throughout the crystallization process for analysis purposes.
- Future Direction: Slow addition of PAE's to a constant NaCl solution is another assembly variable to be considered further.



Phase Separation



Backflow



Change in Surrounding Lighting

Thank You!

